

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – Industry Problem Solving Task

Course Code:	CSA06	Course Title:	Design and Analysis of Algorithms
CO Assessed:	CO4: Articulation (BL4, BL5)	Assessment:	Assessment Tool 2 – Industry Problem Solving Task
Weightage:	20% of CIA	Total Marks:	100
CO4 Statement:	Solve optimization problems using dynamic programming and greedy strategies.		
Student Name:	AKSHAYA KUMAR CHANDRASEKAR	Reg. No.:	192425350
Section / Batch:	B.TECH AIML	Date:	21-08-2026

Instructions to Students

- Answer the following question. Total Marks: 100.
- Carefully analyze the given questions.
- Explain in detail with sample code if needed.
- Clearly identify all key issues.
- Provide a thorough analysis with validated results and performance evaluation.

Question Type	Focus Area	Questions
Industry Problem Solving Task	Focus on Solving optimization problems using dynamic programming and greedy strategies	Q1

SECTION A – Industry Problem Solving Task

ASSESSMENT-CO4/AT2-Industry Problem Solving Task

[← Back](#)Language: Python C C++ Java

QUESTIONS

Q1

Knapsack – Industry Resource
Allocation
100 marks

1/1 answered

Q1 Knapsack – Industry Resource Allocation

100 marks

A cloud service provider must allocate limited server memory (100 GB) among multiple client applications, each having different memory requirements and revenue values. Analyze the problem and explain how the 0/1 Knapsack algorithm can be applied to maximize total revenue. Design a feasible solution using dynamic programming and justify the selection of this approach. Discuss how overlapping subproblems and optimal substructure are utilized. Suggest suitable tools or platforms for implementation. Analyze the performance in terms of time and space complexity and interpret the results in an industry context.

Your Code

```
1 def knapsack_01(weights, values, W):
2     n = len(weights)
3
4     dp = [[0] * (W + 1) for _ in range(n + 1)]
5
6     keep = [[False] * (W + 1) for _ in range(n + 1)]
7
8     for i in range(1, n + 1):
9         w = weights[i - 1]
10        v = values[i - 1]
11        for cap in range(W + 1):
```

Input

```
4 100
20 30 40 50
100 200 300 400
```

Output

```
700
```

Code of Conduct

/AKSHAYA KUMAR CHANDRASEKAR (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: **AKSHAYA KUMAR CHANDRASEKAR**

RUBRICS FOR CALCULATION

Industry Problem Solving Task

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Industry Problem	20%	Comprehensive understanding of real-world problem and constraints	Good understanding with minor gaps in requirements	Basic understanding; some requirements unclear	Poor understanding; misinterprets problem context
Application of Engineering Concepts	25%	Applies advanced and relevant concepts effectively to solve the problem	Applies appropriate concepts with minor errors	Limited application of concepts	Incorrect or no application of concepts
Solution Design	25%	Innovative, practical, and feasible solution aligned with industry standards	Logical and feasible solution with minor gaps	Basic solution with limited feasibility	Unclear or impractical solution
Use of Tools	15%	Effective use of modern tools, technologies, and real data	Appropriate use of tools with correct implementation	Limited or inconsistent use of tools	Minimal or incorrect use of tools
Analysis	10%	Thorough analysis with validated results and performance evaluation	Good analysis with reasonable validation	Basic analysis with limited validation	Poor or no analysis
Professionalism	5%	Highly professional, well-documented, and clearly presented work	Good documentation and presentation	Basic documentation with some gaps	Poor documentation and presentation

Marks Evaluation:

Criteria	Wt.	Level 4 –	Level 3 –	Level 2 –	Level 1 –
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		Exemplary	Proficient	Developing	Beginning
Understanding of Industry Problem	20%				
Application of Engineering Concepts	25%				
Solution Design	25%				
Use of Tools	15%				
Analysis	10%				
Professionalism	5%				
Total Marks (100):					

Faculty In-charge	Course Coordinator	Head of Department
Name:	Name:	Name:
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____